

NICOLE ERIN KEFFER

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EDUCATION

UNIVERSITY OF COLORADO BOULDER

Bachelor of Science Information Science, m. Economics, Certificate in User Experience

Exp. May 2026

Honors - Dean's List Scholar

CERTIFICATIONS: Social Behavioral Research - Citi Program: Human Research

WORK EXPERIENCE

ATS STUDENT HARDWARE LAB ADMINISTRATOR

October 2025 - Present

University of Colorado Boulder - Office of Information Technology

- **Co-developed an automated dashboard script** to track **lab status and equipment monitoring data**, reducing **manual reporting** and improving **operational visibility** across campus labs.
- Leveraged **operational data analysis** to identify **recurring infrastructure issues**, informing **data-driven workflow improvements and process optimization**.
- **Increased lab monitoring coverage by 200%** through improved **data tracking and reporting processes**, strengthening **equipment reliability metrics**.
- **Redesigned lab walkthrough documentation and tracking workflows**, improving **data capture, reporting consistency, and team coordination**.

PROJECT COORDINATOR & SYSTEMS ARCHITECT

August 2025 - December 2025

University of Colorado Boulder - Digital Legacy Clinic (Jed Brubaker, PhD)

- Designed and documented an **End-of-Life (EOL) Standard Operating Procedure (SOP)** for **digital asset management**, creating **standardized workflows** that improved **onboarding efficiency** and ensured **consistent operational processes** across the team.
 - Conducted **comparative platform analysis and structured research synthesis** on digital legacy systems and account recovery infrastructures.
 - Served as **primary client liaison**, translating **user requirements into structured documentation, system workflows, and project deliverables**.
 - Produced **analytical case studies of digital asset scenarios and platform solutions**, converting research findings into **decision-support documentation and training materials**.
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RESEARCH & DATA PROJECTS

OUR BLUE PLANET: MARINE PROTECTED AREAS ANALYSIS - Python, Altair, Datawrapper

- Analyzed ReefCheck and NOAA datasets to evaluate ecological outcomes of Hawaii marine protected areas (MPAs).
- Performed data cleaning, aggregation, and exploratory analysis in Python to identify trends in reef health and biodiversity indicators.
- Built interactive visualizations and scrollytelling analysis using Altair and Datawrapper to communicate ecological and policy insights.

GALAXY MORPHOLOGY CLASSIFIER & CAPTION GENERATOR - Python, TensorFlow, CNN, LSTM

- Engineered deep learning pipeline to classify galaxy images and generate captions using CNN and RNN architectures from astronomical datasets.
 - Trained and evaluated models in TensorFlow, applying convolutional feature extraction and sequence modeling for image-to-text generation.
 - Implemented data preprocessing and model evaluation workflows to support automated galaxy morphology classification and prediction tasks.
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TECHNICAL SKILLS

Languages: Python, SQL, R.

Data Analysis & Machine Learning: Pandas, NumPy, Scikit-learn, Neural Networks (RNNs), Decision Trees, Random Forests, Predictive Analytics, Data Processing, Feature Engineering.

Data Visualization & BI: Tableau, Power BI, Altair, Matplotlib, Seaborn, Dashboard Development.

Analytics & Methods: Data Cleaning, Data Wrangling, Statistical Analysis, A/B Testing, KPI Development, Quantitative Analysis, Social Network Analysis.